BIO 204
Midterm Past Papers

1. Growth → Change in **(state/number/ size/volume) All**

2. _____ is defined as the time required for cell mass to double. **Doubling time (td)**

3. In Xt = xo eμt, Xo = original biomass concentration

4. Specific growth rate is independent of substrate concentration as long as excess substrate is present.

5. Multi blade impeller is discovered by Jackson (1958)

6. Parts of Fermenter are_____ **Detachable stirrer motor, Vessel pH/O2 Electrodes Exhaust Gas Condenser Dialysis Unit (All)**

7. There is no universal_____ **Bioreactor**

8. The function of the fermenter or bioreactor is to provide a suitable_____ **(Environment)**

9. Nitryl or butyl rubbers are normally used for these_____ **(Seals.)**

10. If medium is fed continuously to such a culture at a suitable rate_____ is achieved eventually, that is, formation new biomass by the culture is balanced by the loss cells from the vessel. **(Steady state)**

11. If qp is strictly growth related then it will change as μ with D and, thus, the product concentration remain **(Constant)**

12. Pirt _____described the kinetics of fixed volume fed-batch culture as follows. **(1979)**

13. Consider a batch culture in which the growth of the process organism_____ the limiting substrate to a limiting level. **(Depleted)**

13. μmax =is the maximum Specific Growth Rate per unit cell mass

14. The kinetics of _____volume in fed-batch culture have been developed by Dunn and Mor (1975) and Pirt (1974, 1975, 1979). **(Variable)**

15. An _____type of continuous culture to the chemostat is the turbidostat. **(Alternative)**

16. What increase the turbulence of fermenter_____ **(Vessel)**

17. Axial flow impellers provide gentler but _____mixing and are used for reactions involving shear sensitive cells and particles. **(Efficient)**

18. The same medium used to establish the batch culture is added, resulting in an increase in_____ **(Volume)**

VU Medical Zone Effort by Amaan Khan

19. AISI Grade 316 steels which contain 18% chromium, 10% nickel and 2-2.5% molybdenum are now com mainly_____ in fermenter construction. **(Used)**

20. Multi blade impeller is discovered by Jackson_____ **(1958)**

21. Storage phase is____ **(Equivalent to the late exponential phase)**

22. Grade 316 steel with _____% chromium and_____ % molybdenum of other material are use in construction of fermenter. **(18% and 2-2-5%)**

23. Pressure of air compressed in fermenter_____ **(1.5 to 3.0 atmospheres)**

24. The agitation consists of _____ and baffles.

25. In the equation Xt=X.eumax, original biomass is_____ **(X_0)**

26. If qp is strictly growth related _____concentration will rise. **(Product)**

27._____flow is more efficient **(Axial)**

28._____controls the liquid flow in fermenter **(Baffles)**

29. No_____ fermenter is present. **(Required fermenter should be designed)**

VU Magical Zone Effort by Amaan Khan

SUBJECTIVE

2 marks

Qno1: Difference between quasi steady state and steady state of chemostat.

The major difference between the steady state of a chemostat and the quasi steady state of a fed-batch culture is that μ is constant in the chemostat but decreases in the fed batch.

Qno2: Difference between Axial flow and Radial flow.

With radial flow mixing, the liquid flow from the impeller is initially directed towards the wall of the reactor; i.e. along the radius of the tank.

With axial flow mixing, the liquid flow from the impeller is directed downwards towards the base of the reactor, i.e. in the direction of the axis of the tank.

Qno3: Name the four accessories of fermentation process.

• Baffle

• Sterile compressed air (at 1.5 to 3.0 atmospheres)

• Chilled water (12 to 15°C) • Cold water (4°C)

• Hot water

Qno4: covert Monod equation into linear.

To convert Monod equation into Linear equation we can write it as: 1/μ=1/ μmax+Ks /μmaxS Y=a+bx .The plot of 1/μ against 1/S produces a straight line with intercept on the y axis 1/ μmax and slope equalsto: **Ks / μmax.**

Qno5: What is bottom agitators and top agitator?

Bottom entry agitators tend to require more maintenance than top entry impellers due to the formation of crystals and other solids in the seals.

Qno6: Define Fed -batch culture?

Yoshida et al., (1973) introduced the term fed-batch culture to describe batch cultures which are fed continuously, or sequentially, medium, with without the removal of culture fluid.

Qno7: Why glass use in fermenter?

Glass is useful because it gives smooth surfaces, is non-toxic, corrosion proof and it is usually easy to examine the interior of the vessel.

Qno8: What is the appropriate containment requirements can be applied for hazardous groups?

Effort by Amaan Khan

Procedures	GILSP*	Containment Category		
		1	2	3
Written instructions and code of practice	+	+	+	+
Biosafety manual	–	+	+	+
Good occupational hygiene	+	+	+	+
Good Microbiological Techniques (GMT)	–	+	+	+
Biohazard sign	–	+	+	+
Restricted access	–	+	+	+
Accident reporting	+	+	+	+
Medical surveillance	–	+	+	+

*Unless required for product quality,–, not required; +, required.

EFFORT BY AMAAN KHAN

Procedures	GILSP*	Containment Category		
		1	2	3
Primary containment: Operation and equipment				
Work with viable micro-organisms should take place in closed systems (CS), which minimize (m) or prevent (p) the release of cultivated micro-organisms	–	m	p	p
Treatment of exhaust air or gas from CS	–	m	p	p
Sampling from CS	–	m	p	p
Addition of materials to CS, transfer of cultivated cells	–	m	p	p
Removal of material, products and effluents from CS	–	m	p	p
Penetration of CS by agitator shaft and measuring devices	–	m	p	p
Foam-out control	–	m	p	p

m, minimize release. The level of contamination of air, working surface and personnel shall not exceed the level found during microbiological work applying Good Microbiological Techniques.
p, prevent release. No detectable contamination during work should be found in the air, working surfaces and personnel.

VU Medical Zone Effo

EFFORT BY AMAAN KHAN

Procedures	GILSP*	Containment Category		
		1	2	3
Secondary containment: Facilities				
Protective clothing appropriate to the risk category	+	+	+	+
Changing/washing facility	+	+	+	+
Disinfection facility	–	+	+	+
Emergency shower facility	–	–	+	+
Airlock and compulsory shower facilities	–	–	–	+
Effluents decontaminated	–	–	+	+
Controlled negative pressure	–	–	–	+
HEPA filters in air ducts	–	–	+	+
Tank for spilled fluids	–	–	+	+
Area hermetically sealable	–	–	–	+

* Unless required for product quality,–, not required; +, required.

VU Medical Zone

Qno9: What is chemostate?

The growth of the cells in a continuous culture of this type is controlled by the availability of the growth limiting chemical component of the medium and, thus, the system is described as a chemostat.

Qno10: How can we calculate both live and dead cells?

Cell growth is also measured by counting total cell number of the microbes present in that sample. Total cells (both live and dead) of liquid sample are counted by using a Haemocytometer/Neubauer Chamber.

If there is dilute culture, direct cell counting can be done. However, the cell culture of high density can be diluted. Otherwise clumps of cells would be formed which would create problem in exact counting of bacterial cells.

Qno11: Can results of chemostat be same with foregoing theory?

No, the results of chemostat experiments may differ from those predicted by the foregoing theory. The reasons for these deviations may be anomalies associated with the equipment or the theory not predicting the behavior of the organism under certain circumstances.

Qno12: Give two defects of given by Collins.

Risk such as those given by Collins (1992):

EFFORT BY AMAAN KHAN

1. The known pathogenicity of the micro-organism.

2. The virulence or levels of pathogenicity of the microorganisms are the diseases it causes mild or serious?

3 marks

Qno1: Describe Scale of Fermenter/Bioreactor in our daily life.

• Lab-scale Fermenters Ex-situ sterilization but seed fermenter is not required for inoculums development, slants or flask cultures can be utilized.

• Pilot-scale Fermenters In-situ sterilization but seed fermenter is not required for inoculums development, slants or flask cultures can be utilized.

• Industrial-scale Fermenters In-situ sterilization and seed fermenter is required for inoculums development.

Qno2: What is the situation of quasi steady state?

Thus, (ds/dt) ≃ 0, Although the total biomass in the culture (X) increases with time, cell concentration (x) remains virtually constant, that is (dx/dt) ≃ 0 and therefore μ ≃ D. This situation is termed a quasi-steady state.

Qno3: Drive the Equation for the Fed-batch fermenter.

If the limiting substrate is then added in a concentrated feed such that the broth volume remains almost constant then: dx/dt =GY where and G is the substrate feed rate (g dm-3h-1) Y is the yield factor. But dx/ dt =μx, thus substituting for dx/ dt in equation (2.34) gives: μx= GY, and thus: μ= GY/x

Qno4: Relation of qp and M (mew) in fed-batch culture.

Product concentration in a fed-batch system over the time of the fermentation will be dependent on the relationship between qp and μ (hence D). If qp is strictly growth related then it will change asμ with D and, thus, the product concentration remain constant. However, if qp is constant and independent of μ, then product concentration will the start of the cycle when Dp is greater than qpx, but will rise with time as D decreases and qpx become greater than Dp.

Qno5: What should be controlled in bioreactor In general?

There is no universal bioreactor. The general requirements of the bioreactor are as follows:

1. The vessel should be capable of being operated aseptically for a number of days and should be reliable in long-term operation and meet the requirements of containment regulations.

2. Adequate aeration and agitation should be provided to meet the metabolic requirements of the micro-organism. However, the mixing should not cause damage to the organism.

3. Power consumption should be a slow as possible.

4. A system of temperature control should be provided.

EFFORT BY AMAAN KHAN

VU Medical Zone Effort by Amaan Khan

5. A system of pH control should be provided.

6. Sampling facilities should be provided.

Qno6: Why the bottom entry agitators tend to require more maintenance than top entry impellers?

Bottom entry agitators tend to require more maintenance than top entry impellers due to the formation of crystals and other solids in the seals.

Qno7: G and Y stands for in the equation Dt/dt=GY.

G is the substrate feed rate (g dm^{-3} h^{-1})

Y is the yield factor

5 marks

Qno1: What is Agitators radial flow and axial flow characteristics?

❑Agitators are classified as having radial flow or axial flow characteristics.

❑With radial flow mixing, the liquid flow from the impeller is initially directed towards the wall of the reactor; i.e. along the radius of the tank.

❑With axial flow mixing, the liquid flow from the impeller is directed downwards towards the base of the reactor, i.e. in the direction of the axis of the tank.

❑Radial flow impellers are primarily used for gas-liquid contacting (such as in the mixing of sparged bioreactors) and blending processes.

❑Axial flow impellers provide gentler but efficient mixing and are used for reactions involving shear sensitive cells and particles.

Qno2: Write the major difference between steady state of the chemostat and quasi steady state of fed batch culture? Also describe the effects of the following values U, umax, Ks on biomass.

The major difference between the steady state of a chemostat and the quasi steady state of a fed-batch culture is that μ is constant in the chemostat but decreases in the fed batch.

The effect of dilution rate on the steady-state biomass and residual substrate concentrations in a chemostat of a microorganism with a low Ks value for the limiting substrate, compared with the initial substrate concentration.

Figure in previous slide illustrates the continuous culture behavior of a hypothetical bacterium with a low Ks value for the limiting substrate, compared with the initial limiting substrate concentration. With increasing

VU Medical Zone Effort by Amaan Khan

dilution rate, the residual substrate concentration increases only slightly until D approaches µmax when s increases significantly. The dilution rate at which x equals zero (that is, the cells have been washed out of the system) is termed the critical dilution rate (D crit) and is given by the equation: D crit = µmax SR / (Ks+SR).

The kinetic characteristics of an organism (constants Y, µmax and Ks. The value of Y affects the steady-state biomass concentration; the value of µmax affects the maximum dilution rate that may be employed and the value of Ks affects the residual substrate concentration (and, hence, the biomass concentration) and also the maximum dilution rate that may be used.

VU Medical Zone Effort by Amaan Khan

EFFORT BY AMAAN KHAN